Central University of Jammu
Hackathon 2026

- Problem Statement - Development of Piezoelectric Sensor for Detection Of Microplastics in water samples.
- Category-Hardware
- Team Name- Team Diamonds

TEAM
DIAMONDS

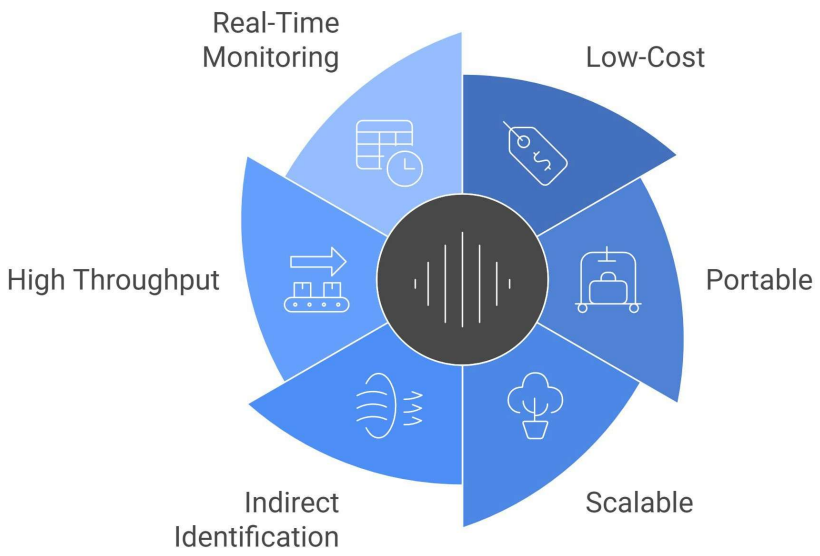
Smart Acoustic Sensor System for Real-Time Microplastic Detection in water sample

TEAM DIAMONDS

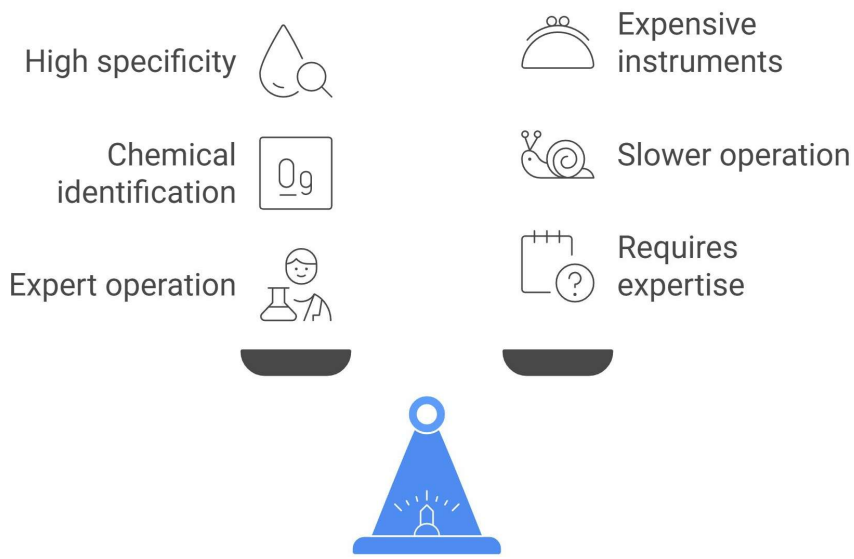
An Innovative Approach Using Piezoelectric Transducers, ESP32, and Machine Learning.

For the detection of microplastics in the sample using the acoustics signatures of the microparticle present in the sample.

Acoustic System Overview (Our Project)

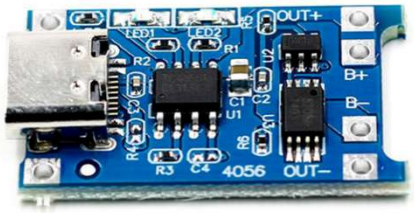


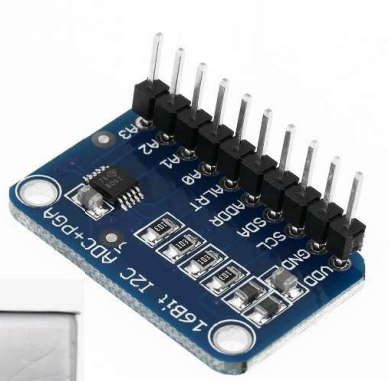
Spectroscopic Analysis



What are the components we are going to use ?

- ESP32
- piezo-electric sensors
- Acrylic chamber
- ADC module
- Cables
- A pretrained dataset
- Battery
- Capacitors
- Charging module

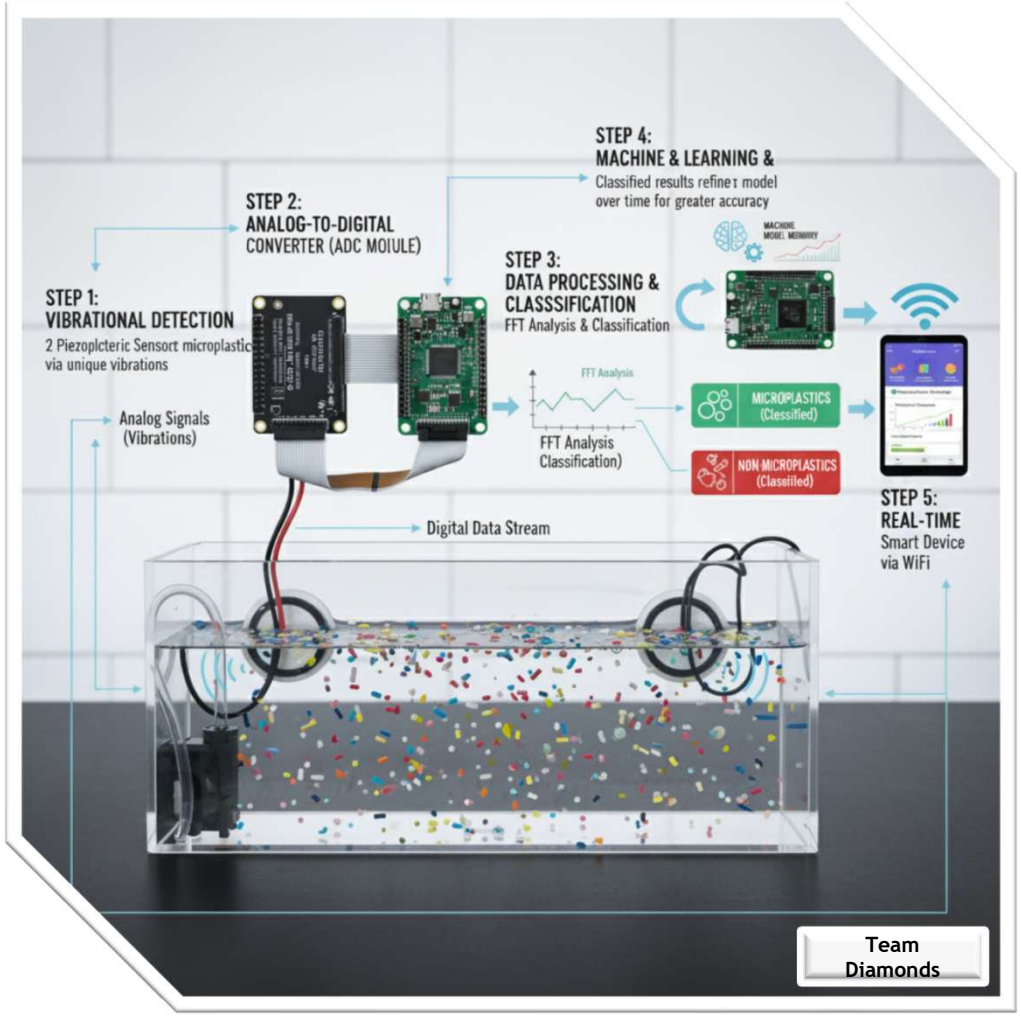




TECHNICAL APPROACH

WORKFLOW

1. We will use 2 piezoelectric sensors for detecting microplastics in the sample using the vibrational signature.
2. Then the analog signal intercepted by the sensors are transferred to the ADC module which converts them from analog to digital.
3. Which is further transferred to our ESP32 module,which processes the data and perform FFT on the Data.
4. Therefore classifying the detected objects as microplastics and non microplastics.
5. These results are stored in our models memory which by using ML refines itself with time for more accurate results in future.
6. These results are reflected in a design dashboard on a smart device in real time using wifi transmission.



Sample Code for Explanation

- This is the Link to the Source file for explaining the workflow of our project :workflow\Workflow.cpp
- At the backend it will pass the values to the Web portal which we are implementing as IOT in our model.
- This will show the data in real-time to the user.
- A sample of this attached programs real-time output is shown below: workflow\output.html

FEASIBILITY AND VIABILITY

Feasibility

Low cost: Uses ESP32, ADC, and piezoelectric sensor — cheap and easy to find.

Simple design: ESP32 can handle real-time FFT without extra hardware.

Eco-friendly: Acoustic sensing avoids chemicals, safe for the environment.

Scalable: Can be miniaturized for compact and portable use.

Integration: Works with Wi-Fi/Bluetooth for IoT connection.

Viability

Affordable solution: Lower cost than spectroscopy- or microscopy-based detection methods.

Energy-efficient: Low power consumption allows battery or solar operation.

Field deployable: Portable chamber design enables in-situ testing of water samples.

Potential for citizen science: Simple design can be distributed for large-scale monitoring.

Future adaptability: Can add AI/ML for better accuracy and faster classification.

COMMERCIAL ASPECT

- Target audience : Environmental Researchers & Scientists, Government & Regulatory Bodies, NGOs & Environmental Organizations, Industries & Water Treatment Plants.

- In the further prototypes we can implement acoustic sweeping in the chamber, so that we can not only detect but purify the water body but currently we are targeting the audience which mainly focuses on handling health hazards.

- Our prototype's estimated cost is from INR 3000 to INR 5000.for making our prototype which will maintain the efficiency with cost.

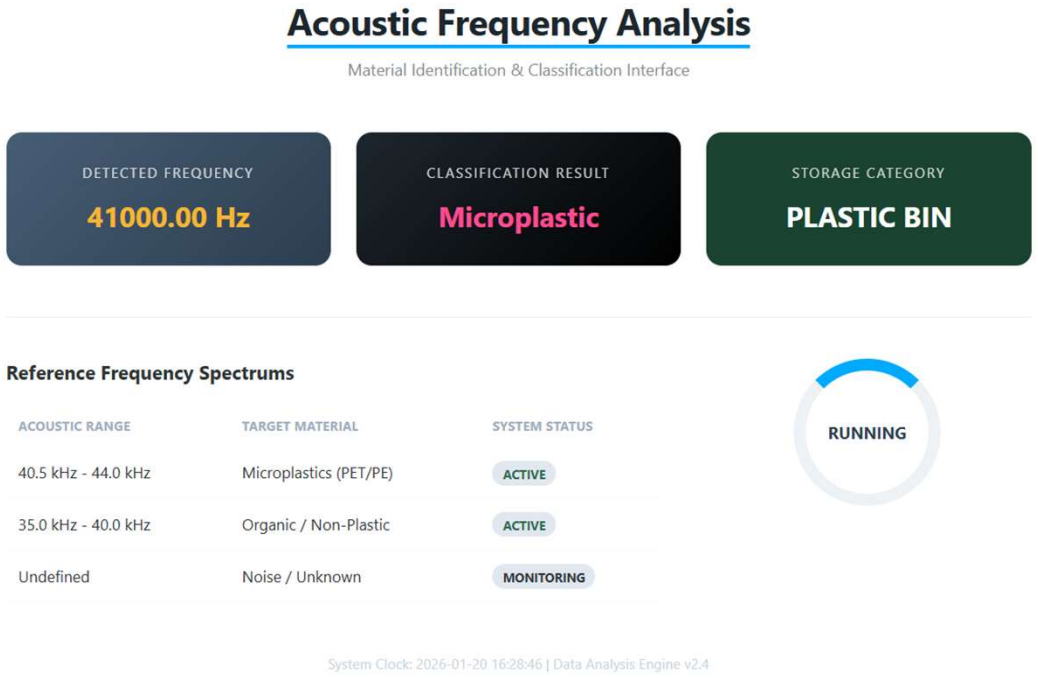
IMPACT AND BENEFITS

Potential impact on the target audience

- The project's high potential lies in enabling **decentralized, real-time microplastic monitoring**, offering a faster, cost-effective solution for researchers and policymakers.

❖ Benefits of the solution:

1. Cost-Effectiveness
2. Real-Time & Rapid Results
3. Portability & Scalability



RESEARCH AND REFERENCES

□ Here are some research and reference papers our project:

- https://www.researchgate.net/publication/381808327_Research_progress_on_piezoelectric_acoustic_transducers_Principles_materials_performance_and_applications
- https://www.researchgate.net/publication/337156914_Acoustic_focusing_of_microplastics_in_microchannels_A_promising_continuous_collection_approach
- https://www.mdpi.com/1424-8220/21/10/3532
- **Google books**
- **AI-powered chatbots**